8.Perform basic Image Handling and processing operations on the image is to read an image in python and Dilate an Image using Dilate function

Aim

To write a Python program to read an image and perform image dilation using the cv2.dilate() function in OpenCV.

Procedure

1. Import OpenCV and NumPy libraries.
2. Read the input image in grayscale.
3. Convert the image into binary format using thresholding.
4. Create a structuring element or kernel.
5. Apply dilation using cv2.dilate().
6. Display the original binary image and dilated image.

Code

```
import cv2
import numpy as np

img = cv2.imread("sample.jpg", 0)

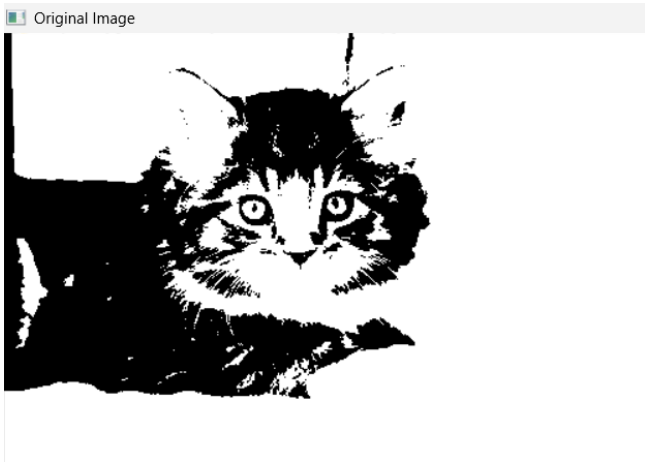
if img is None:
    print("Image not found")
else:
    _, binary = cv2.threshold(
        img, 127, 255, cv2.THRESH_BINARY
    )

    kernel = np.ones((5, 5), np.uint8)
    dilated = cv2.dilate(binary, kernel, iterations=1)

    cv2.imshow("Original Image", binary)
    cv2.imshow("Dilated Image", dilated)
```

```
cv2.waitKey(0)
cv2.destroyAllWindows()
```

SAMPLE INPUT:



SAMPLE OUTPUT:



Result

Thus, image dilation was successfully performed using the cv2.dilate() function. The white regions in the binary image became larger.